

1C-3**a)**

$$\begin{vmatrix} 1 & 2 \\ 1 & -1 \end{vmatrix} = -3$$

So the area of the parallelogram is 3. Area of the triangle is $\frac{3}{2}$.

b)

Sides are

$$PQ = (0, -3), \quad PR = (1, 1)$$

$$\begin{vmatrix} 0 & -3 \\ 1 & 1 \end{vmatrix} = 3$$

So area of the parallelogram is 3, Area of the triangle is $\frac{3}{2}$.

1C-4

$$\begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ x_1^2 & x_2^2 & x_3^2 \end{vmatrix} = x_2x_3^2 + x_3x_1^2 + x_1x_2^2 - x_2^2x_3 - x_3^2x_1 - x_1^2x_2$$

$$= (x_1 - x_2)(x_1 - x_3)(x_2 - x_3)$$

1C-5**a)**

$$\begin{vmatrix} x_1 & y_1 \\ x_2 + ax_1 & y_2 + ay_1 \end{vmatrix} = x_1y_2 + ax_1y_1 - x_2y_1 - ay_1x_1$$

$$= \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$$

b) is similar.**1C-7**

The heads of two vectors are on the unit circle. The area of the parallelogram they span is maximum when the vectors are perpendicular.

$$ab \sin \theta = 1 \cdot 1 \cdot \sin \theta$$

Maximum when

$$\theta = \frac{\pi}{2}$$

Hence the maximum value of

$$\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$$

is 1 (area of the unit square).

1C-9

$$PQ = (0, -1, 2), \quad PR = (0, 1, -1), \quad PS = (1, 2, 1)$$

Volume of the parallelepiped:

$$V = \begin{vmatrix} 0 & -1 & 2 \\ 0 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix}$$

$$V = \pm 4$$

Volume of the tetrahedron:

$$V = \frac{1}{6} \begin{vmatrix} 0 & -1 & 2 \\ 0 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix} = \frac{4}{6}$$

$$V = \frac{2}{3}$$

1C-10

Thinking of them as origin vectors, A lies in the plane of B and C .

Therefore the volume of the parallelepiped spanned by the three vectors is

$$V = 0$$

1D-1

a)

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & 1 \\ 2 & -1 & -1 \end{vmatrix} = 3\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}$$

b)

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & -3 \\ 1 & 1 & -1 \end{vmatrix} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$$

1D-2

$$PQ = i + j - k, \quad PR = 3i + j - 2k$$

$$PQ \times PR = \begin{vmatrix} i & j & k \\ 1 & 1 & -1 \\ 3 & 1 & -2 \end{vmatrix} = -i + 5j + 4k$$

Area of the triangle:

$$\text{Area} = \frac{1}{2}|PQ \times PR|$$

1D-5

We get a third vector (properly oriented) perpendicular to A and B by using $A \times B$.

a)

$$\sin \theta = 1 \Rightarrow \theta = \frac{\pi}{2}$$

The two vectors are orthogonal.

b)

$$|A||B|\sin \theta = |A||B|\cos \theta$$

Since $\tan \theta = 1$

$$\theta = \frac{\pi}{4}$$

1D-3

$$A \times B = \begin{vmatrix} i & j & k \\ 2 & -1 & 0 \\ 1 & 2 & 1 \end{vmatrix} = -i - 2j + 5k$$

Make these unit vectors:

$$i' = \frac{A}{\sqrt{5}}, \quad j' = \frac{B}{\sqrt{6}}, \quad k' = \frac{i - 2j + 5k}{\sqrt{30}}$$

1D-4

$$(i \times i) \times j = 0 \times j = 0$$

$$i \times (i \times j) = i \times k = -j$$

1D-6

Taking the cube so P is at the origin and three coterminous edges are $\mathbf{i}, \mathbf{j}, \mathbf{k}$, the three diagonals of the faces are

$$\mathbf{i} + \mathbf{j}, \quad \mathbf{j} + \mathbf{k}, \quad \mathbf{i} + \mathbf{k}$$

Volume of parallelepiped by diagonals:

$$\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} = 2$$

1D-7

We have

$$PQ = (-2, 1, 1), \quad PR = (-1, 0, 1), \quad PS = (2, 1, -2)$$

Volume of parallelepiped:

$$\begin{vmatrix} -2 & 1 & 1 \\ -1 & 0 & 1 \\ 2 & 1 & -2 \end{vmatrix}$$

Volume of tetrahedron:

$$V = \frac{1}{6} \begin{vmatrix} -2 & 1 & 1 \\ -1 & 0 & 1 \\ 2 & 1 & -2 \end{vmatrix}$$

1D-8

One determinant has rows in the order A, B, C , the other represents $C \cdot (A \times B)$ and therefore has its rows in the order C, A, B .

To change the first determinant into the second, interchange the second and third rows, then the first and second rows. Each interchange multiplies the determinant by -1 .

1D-9

a) Lift the triangle up into the plane $z = 1$ so its vertices are at the three points

$$P_i = (x_i, y_i, 1), \quad i = 1, 2, 3$$

Volume of tetrahedron $OP_1P_2P_3$

$$\begin{aligned} V &= \frac{1}{3}(\text{height})(\text{base}) \\ &= \frac{1}{3}(1)(\text{area of triangle}) \end{aligned}$$

Also,

$$V = \frac{1}{6}(\text{volume of parallelepiped spanned by } OP_i)$$

$$= \frac{1}{6} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Therefore,

$$\text{Area of triangle} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

b) Subtraction of the first row from the second and the first from the third does not change the value of the determinant.

$$\begin{aligned} \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} &= \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 - x_1 & y_2 - y_1 & 0 \\ x_3 - x_1 & y_3 - y_1 & 0 \end{vmatrix} \\ &= \frac{1}{2} \begin{vmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{vmatrix} \end{aligned}$$

IE: Lines and Planes

IE-1

a)

$$(x - 2) + 2y = 2(z + 1)$$

$$x + 2y - 2z = 4$$

b)

$$N = (1, 1, 0) \times (2, -1, 3)$$

$$N = \begin{vmatrix} i & j & k \\ 1 & 1 & 0 \\ 2 & -1 & 3 \end{vmatrix} = 3i - 3j - 3k$$

Removing the common factor 3:

$$x - y - z = 0$$

c)

$$PQ = (1, -1, 1), \quad PR = (-2, 3, 1)$$

$$N = PQ \times PR$$

$$N = \begin{vmatrix} i & j & k \\ 1 & -1 & 1 \\ -2 & 3 & 1 \end{vmatrix} = -4i - 3j + k$$

Equation through point $P(1, 0, 1)$:

$$-4(x - 1) - 3y + (z - 1) = 0$$

d)

$$-4x - 3y + z = 3$$

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$